BhumiSetu

India's Land Decision Infrastructure

Master Product Requirements Document

Parts 16 to 20: UX, Security, Analytics, Roadmap & Appendices

Document Grade: Investor & Engineering Ready

Date: July 2026

PART 16: UX SYSTEM & DESIGN LANGUAGE

BhumiSetu operates in a high-anxiety, low-trust environment. The UI must be the antidote to this anxiety. We employ a "Trust Architecture" design philosophy—clean, highly structured, and institutional.

1. Visual Language

- **Typography:** 'Inter' for UI elements (highly legible, objective) and 'Geist' for marketing/headings to add a premium, tech-forward feel.
- **Color Palette:**
  - Primary: *Bhumi Emerald* (#10b981) - Signifies safety, approval, and verification.
  - Surface/Background: Minimalist off-whites (#f8fafc) to make data pop.
  - Text: Slate grays (#0f172a to #64748b) for extreme readability.
- **Spacing:** Strict 8pt grid system. Dense data views (Enterprise Portal) use 4pt micro-spacing.

2. Components & Animations

- **Status Badges:** Never use ambiguous colors. Red (Dispute/Risk), Yellow (Pending Human Review), Green (Verified).
- **Animations:** No bouncy or playful animations. Use subtle fades and structured skeleton loaders (simulating document scanning) to build anticipation and perceived value of the AI processing.
- **Accessibility (A11y):** WCAG 2.1 AA compliant. High contrast mode for the Partner App (surveyors reading screens in direct sunlight).

PART 17: SECURITY & COMPLIANCE

We are dealing with highly sensitive financial and legal data. Security is a tier-1 product feature.

1. Authentication & Authorization

- **Passwordless Auth:** SMS/WhatsApp OTP via Phone Number. Eliminates password fatigue and ties identity directly to Indian telecom KYC. WebAuthn (FaceID/Fingerprint) enabled for returning mobile users.
- **RBAC (Role-Based Access Control):** Strict separation. A Partner Surveyor can only view the exact coordinates of their assigned task, not the buyer's financial details or the legal document analysis.

2. Data Protection & Compliance

- **DPDP Act Compliance:** Adherence to India's Digital Personal Data Protection Act. Explicit consent flows required before querying government databases on behalf of a user.
- **Audit Trails:** Immutable event logging for every case state change. If an AI summary is overridden by a human lawyer, the timestamp, reason, and user ID are permanently recorded.
- **Encryption:** AES-256 for all PII and document storage at rest (AWS S3/RDS).

PART 18: ANALYTICS & TELEMETRY

We manage what we measure. Telemetry is injected at the component level to track drop-offs and operational bottlenecks.

1. Core Product Metrics (KPIs)

Metric	Definition	Target
TTR (Time-to-Resolution)	Time from case payment to final report generation.	< 5 Days (Tier 2)
Partner Fulfillment Rate	% of survey tasks accepted within 15 minutes of broadcast.	> 90%
AI Automation Rate	% of documents parsed without triggering human exception handler.	> 85%
Enterprise API Latency	Response time for the Digital Health Check endpoint.	< 3 Seconds

2. Funnel Tracking (PostHog/Mixpanel)

- **Retail Conversion Funnel:** Landing Page -> Enter Survey Number -> View Loading State -> View Teaser Report -> Paywall -> Case Initiated.

PART 19: ROADMAP

1. MVP (Months 0-6): The Execution Engine

- Focus: Launch in Maharashtra (highest quality digital land records).
- Deliverables: Customer Web App, Partner Mobile App (Android), AI Document Agent (Marathi/English 7/12 parsing), Manual Legal QA workflow.

2. Phase 2 (Months 7-18): Scale & Enterprise (B2B SaaS)

- Focus: Expand to Karnataka, Telangana, and Gujarat.
- Deliverables: Launch Enterprise Portal & API. Onboard 3 NBFC pilot partners for bulk portfolio underwriting. Fully agentic workflow routing.

3. Phase 3 (Years 2-3): Predictive Intelligence

- Focus: Utilizing our Land Decision Graph to predict disputes before they happen.
- Deliverables: Predictive Risk Scoring (e.g., flagging a parcel because neighboring parcels have active litigations). Subscription monitoring products for remote landowners.

4. The 5-10 Year Vision

BhumiSetu becomes the **default standard for land trust in India**. Banks will not sanction a land loan without a BhumiSetu Trust Score API call. We build the base infrastructure required for the eventual tokenization and liquid trading of Indian real estate.

PART 20: APPENDICES

1. Glossary

- **7/12 Extract:** Revenue document in Maharashtra/Gujarat denoting land ownership, area, and crop details.
- **Encumbrance Certificate (EC):** A record of all registered transactions (sales, mortgages) on a property over a specific period.
- **ULPIN:** Unique Land Parcel Identification Number (Aadhaar for land).

2. Business Rules & Edge Cases

> **Critical Business Rules**
>
> - **No Geofence, No Payout:** Surveyors cannot submit a report unless their device GPS confirms they were within 10 meters of the parcel polygon.
> - **AI Confidence Threshold:** Any extraction with an AI confidence score < 92% is mandatorily routed to the Human Exception Queue.
> - **Zero Refund Policy on Disputed Land:** We charge for the discovery of truth. If a user pays for a diligence report and we uncover massive fraud, the service is fulfilled successfully. We deliver bad news efficiently.

3. Failure Scenarios

- **Scenario:** State Government Land Portal is down for maintenance.
- **Mitigation:** Workflow Agent pauses document scraping, updates Case Status to `Awaiting_Gov_Portal`, and automatically notifies the user via WhatsApp explaining the upstream delay.

End of Document - Master PRD v1.0.0